MOLECULAR BIOLOGY

GENETIC MATERIAL – DNA

Freidrich Meischer (1868), a 22 yr old Swiss physician isolated a substance from the pus cells obtained from discarded surgical bandages. He found that this chemical compound was rich in N_2, P & S. He named it as **nuclein**. His student Richard Allman found that nuclein was of high molecular weight and it also contains acidic nature. So he named it as 'Nucleic acid'.

OCCURENCE:

- Virus
- Prokaryotic cells
- Eukaryotic cells
- Extra-chromosomal DNA(plasmid)
- Organelle DNA (chloroplast and mitochondrial DNA)

DNA as GENETIC MATERIAL

Genetic material is the material composed of genes and genes are units of heredity which controls the expression of character or traits in organisms. It is capable of replicating & being distributed from one generation to next generation.

DIRECT EVIDENCE FOR DNA AS GENETIC MATERIAL

The most conclusive evidence for DNA as the genetic material came from following 3 experiments

- → Griffith transformation expt.
- → Hershey & Chase (bacteriophage expt.)
- → Laderberg & Tatum (bacterial conjugation expt.)

GRIFFITH TRANSFORMATION EXPT

Griffith (1928), a British bacteriologist noted 2 strains of bacterium *Diplococcus pneumoniae* (*Streptococcus pneumoniae*) when grown on nutrient agar medium

Type 1 – Smooth(S) with glistering appearance owing to be virulent.

Type 2 – Rough (R) owing to be non- virulent in nature.

S & R occur in several forms depending on the antigen production. They are S-I, S-II & S-III & R-I, R-II & R-III. Griffith utilized R-II & S-III type for his expt.

- ⇨ When Griffith injected healthy mice with S- strain bacteria, the mice developed pneumonia & died. This suggests that S-strain are virulent.
- ⇨ When R-strain bacteria was injected into mice, they showed no symptoms of illness, indicating its non-virulent nature.
- ⇨ Heat killed S-strain was found to be non-virulent.
- ⇨ Griffith mixed R-strain & heat killed S-strain bacteria & injected them to mice.
- ⇨ The mice died and when observed the blood sample he found live S-strain

Griffith was unable to explain his findings. He said that R-II strain has been transformed to S-strain.

Smooth strain S-II → MICE → DIES

Rough strain R-II → MICE → LIVES

Heat killed S-strain → MICE → LIVES

R-strain + Heat killed S-strain → MICE → DIES

EXPERIMENT CONFIRMATION (OSWALD T. AVERY (1944))

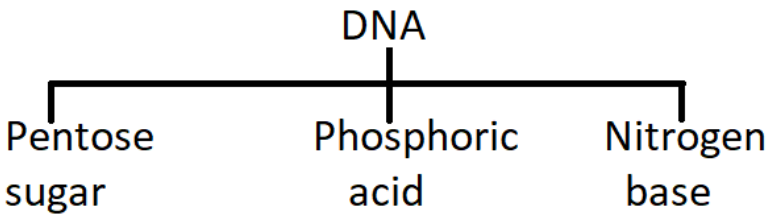
When S-III colonies were grown in agar media the resulting colonies were S-III. When R-II were grown it was R-II. When heat killed S-III & R-II are grown together the stain develops polysaccharide coat around it & give rise to S-III type.

He purified DNA from bacteria of S-strain & performed. No change occurred if DNA was destroyed using DNase enzyme. The change continued if they used other enzymes like RNase, Protease & carbohydrates

Avery noted that newly acquired characteristic of the R-strain namely smooth surface & virulent was passed onto subsequent generation. This process whereby R-strain was converted into S-strain was named transformation

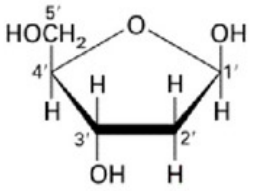
DNA CHEMICAL COMPOSTION

A biochemical analysis of purified DNA shows that it is made of 3 components-



Sugar

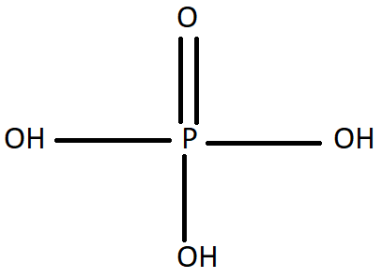
It is a monosaccharide composed of 5 carbon atoms. Hence it is pentose with a chemical formula $C_5H_{10}O_4$.



Deoxyribose

Phosphoric acid

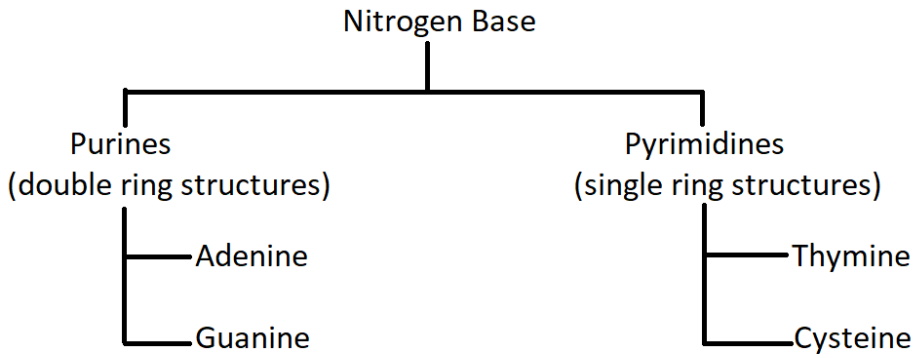
It is ortho-phosphoric acid with molecular formula H_3PO_4. It is responsible for the acidic character of nucleic acid .



NITROGENOUS BASE

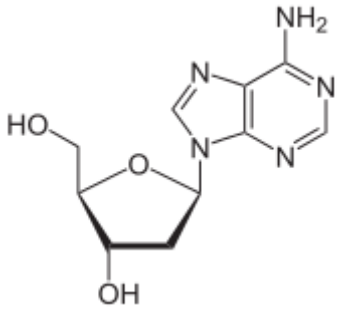
These are N_2 containing organic bases. It is of two main types

1. Purines
2. Pyrimidines



NUCLEOSIDES

A deoxy-ribonucleoside is formed when a molecule of purine or a pyrimidine base is linked to a molecule of the pentose sugar (deoxyribose).

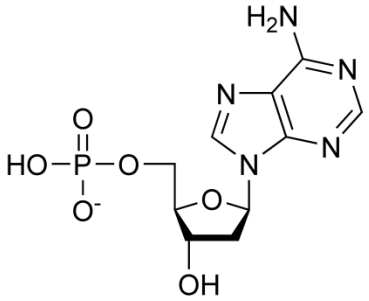


NUCLEOTIDES

Deoxyribose nucleoside + phosphoric acid molecule form deoxy-ribonucleotide.

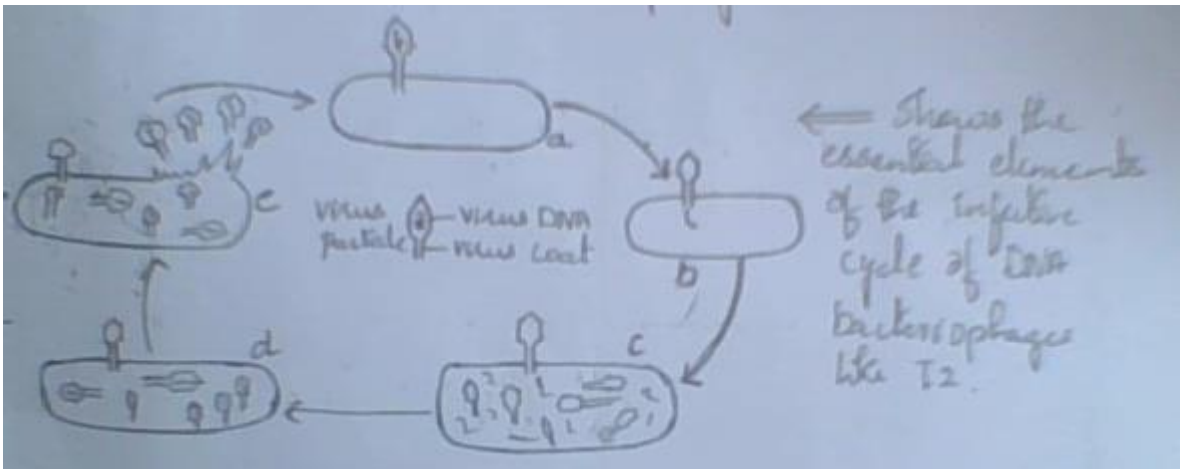
Nucleotides are the building blocks of DNA. When nucleotides are linked to each other to form a chain it is called a polynucleotide strand.

Phosphate attach to 5`C of 1 sugar & 3`C of another sugar this bond is called as phosphodiester bond. Sugar-phosphate-sugar form a backbone to the double-helix structure of DNA.



THE HERSHEY AND CHASE EXPERIMENT

In 1956 (7 yrs after Avery's demonstration that genes were DNA), two geneticists – A.D. Hershey and Martha Chase provided further proof. They worked with a DNA virus called T_2, which infects *E.coli* (so called bacteriophage).



→ The virions attach to the surface of their host cell.
→ The protein of the capsid inject the DNA core into the cell.

- → Once within the cell, some of the bacteriophage genes(the `early` genes) are transcribed (by the host`s RNA polymerase) and translated (by the host`s ribosomes, tRNA, etc.) to produce enzymes that will make many copies of the phage DNA and will turn-off (even destroy) the host`s DNA.
- → As fresh copies of phage DNA accumulate, other genes(the late genes) are transcribed and translated to form the proteins of the capsid.
- → The stockpile DNA cores and capsid proteins are assembled into complete virions.
- → Another `late` gene is transcribed and translated into molecules of lysozyme . the lysozyme attacks the peptidoglycan wall.
- → Eventually the cell ruptures and releases its content of virions ready to spread the infection to new host cell.

Bacteriophages produced within bacteria growing in radioactive culture medium will also be radioactive.

- → If radioactive sulphur atoms(S^{35})are present they will be incorporated into protein coats of the bacteriophages since two of the amino acids –Cysteine and Methionine – contain sulphur, but the DNA will be non-radioactive because there are no sulphur atoms in DNA.
- → If radioactive(P^{32}) is used instead the DNA become radioactive because of its many phosphorus atoms but not the proteins.
- → Hershey and Chase found that – When bacteriophages containing P^{32} radioactive, were allowed to infect non-radioactive bacteria ==++++++, all the infected cells because radioactive and in fact, much of the radioactivity was passed on to the next generation of bacteriophages.
- → However, when the bacteria were infected with bacteriophages labelled with S^{35} and then the virus coats removed (by whirling them in an electric blender) practically no radioactivity could be detected in infected cells.

From these experiments, it was clear that the DNA component of the bacteriophages is injected into the bacterial cell while the protein component remains outside. However, it is the injected component - DNA – that is able to direct the formation of new virus particles complete the protein coats.

So here is the proof that DNA is the genetic material.

DNA REPLICATION

DNA carries the information for making all of the cell`s proteins. These proteins implement all the functions of a living organism and determine the organism`s characteristics. When the cell reproduces, it has to pass all this information on to the daughter cells.

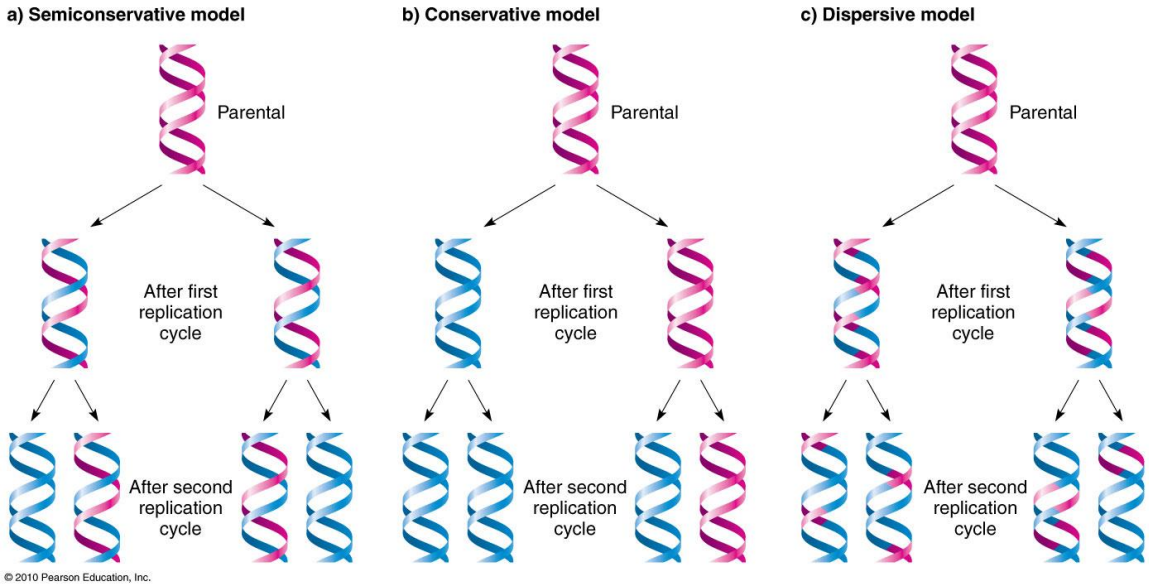
DNA replication is the process of producing two idenical replicas from one original DNA molecule. This biological process occurs in all living organisms and is the basis for biological inheritance.

Different types of cells replicate their DNA at different rates. Some cells constantly divide, like those in your hair and fingernails and bone marrow cells. Other cells go through several rounds of cell division and stop(including specialized cells like those in your brain, muscle and heart.)

Finally, some cells stop dividing, but can be induced to divide, to repair injury (such as skin cells and liver cells). In cells that do not constantly divide, the cues for DNA replication/cell division come in the form of chemicals. These chemicals can come from other parts of the body (hormones) or form the environment.

Three hypothesis had been previously proposed for the method of replication of DNA

1) The ***conservative hypothesis*** proposed that the entire DNA molecule acted as a template for the synthesis of an entirely new one. According to this model, histone proteins bind to the DNA, revolving the strand and exposing the nucleotide bases (which normally line the interior) for hydrogen bonding.
2) In the ***semi conservative hypothesis***, proposed by Watson and Crick, the two strands of a DNA molecule separate during replication. Each strand then acts as a template for synthesis of a new strand.
3) The ***dispersive hypothesis*** is exemplified by a model proposed Max Delbruck, which attempts to solve the problem of unwinding the two strands of the double helix by a mechanism that breaks the DNA backbone every 10 nucleotide or so, untwists the molecule, and attaches the old strand to the end of newly synthesized one. This would synthesize the DNA in short pieces alternating from one strand to the other.

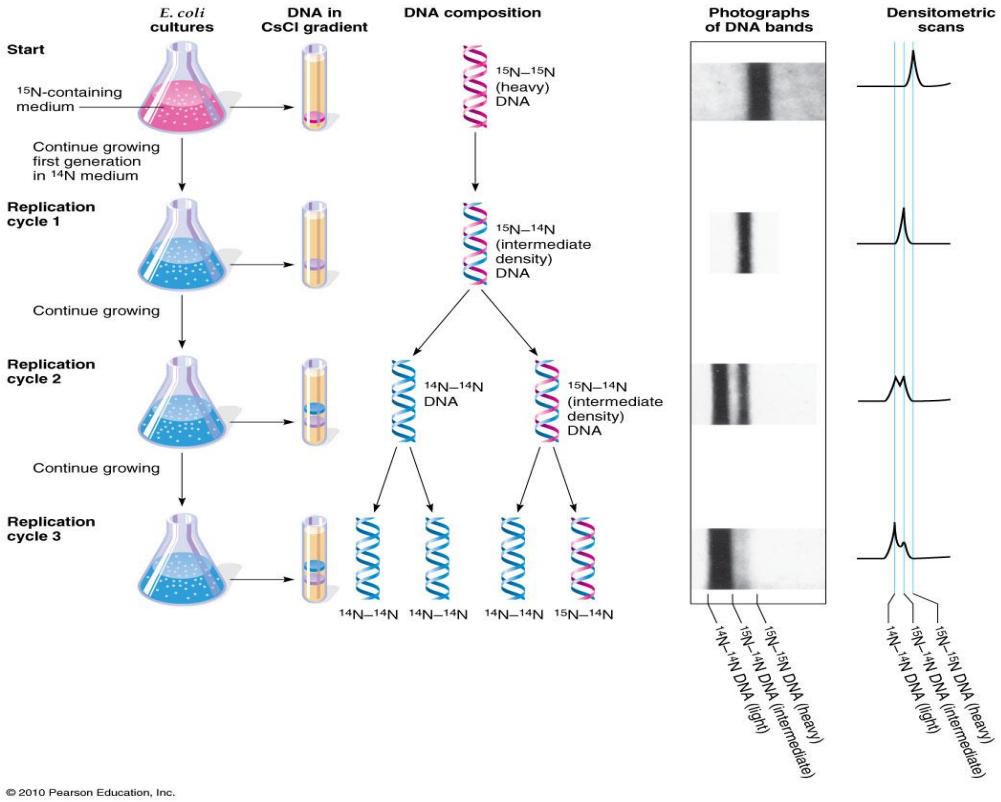


The **Meselson – Stahl experiment** was an experiment by Matthew Meselson and Franklin Stahl in 1958 which supported the hypothesis that DNA replication was semi-conservation. It has been called ***"the most beautiful experiment in biology."***

Experimental procedure and results

1. Nitrogen is a major constituent of DNA. ^{14}N is by far the most abundant isotope of nitrogen, but DNA with the heavier (but non-radioactive) ^{15}N isotope is also functional.
2. *E.coli* were grown for several generations in a medium with ^{15}N. When DNA is extracted from these cells and centrifuged on a salt density gradient, the DNA seperates out at the point at which its density equals that of the salt solution. The DNA of the cells grown in ^{15}N medium had a higher density than cells grown in normal ^{14}N medium.
3. After that, *E.coli* cells with only ^{15}N in their DNA were transferred to a ^{14}N medium and were allowed to divide; the progress of cell division was monitored by microscopic cell counts and by colony assay.
4. DNA was extracted periodically and was compared to pure ^{14}N DNA and ^{15}N DNA. After one replication, the DNA was found to have intermediate density. Since conservative replication would result in equal amounts of DNA of the higher and lower densities (but no DNA of an intermediate density), conservative replication was excluded. However, this result was consistent with both semi conservative and dispersive replication.
5. Semi-conservative replication would result double-stranded DNA with one strand of ^{15}N DNA, and one of ^{14}N DNA, while dispersive replication would result in double-stranded DNA with both strands having, mixtures of ^{15}N and ^{14}N DNA, either of which have appeared as DNA of an intermediate density.
6. The authors continued to sample cells as replication continued. DNA from cells after two replications had been completed was found to consist of equal amounts of DNA with two different densities, one corresponding to the intermediate density of DNA of cells grown for only one division in ^{14}N medium, the other corresponding to DNA from cells grown exclusively in ^{14}N medium.
7. This was inconsistent with dispersive replication, which would have resulted in a single density, lower than the intermediate density of the one generation cells, but still higher than cells grown only in ^{14}N DNA medium, as the original ^{15}N DNA would have been split evenly among all DNA strands.

8. ***The result was consistent with the semi-conservative replication hypot***



Basic rules of replication:

- Semi-conservative
- Starts at the 'origin'
- Can be uni or bi-directional
- Semi-discontinuous
- Synthesis always in the 5-3' direction
- RNA primers required

The mechanism of DNA replication includes

- ***Intiation***
  - proteins bind to DNA and open up double helix
  - Prepare DNA for complementary base pairing
- ***Elongation***
  - Proteins connect the correct sequences of nucleotides into a continuous new strand DNA
- ***Termination***
  - Proteins release the replication complex

Biological significance

- Extreme accuracy of DNA replication is necessary in order to preserve the integrity of the genome in successive generations.
- In eukaryotes, replication only occurs during the phase of the cell cycle.

- Repliction rate in eukaryotes is slower resulting in a higher fidelity/accuracy of replication in eukaryotes

Enzymes used in DNA Replication

Enzyme	Function in DNA replication
DNA Helicase	Also known as helix destabilizing enzyme. Unwinds the DNA double helix at the Replication Fork
DNA Polymerase	Builds a new duple DNA strand by adding nucleotides in the 5` to 3` direction. Also proof reading and error correction.
DNA clamp	A protein which prevents DNA polymerase III from dissociating from the DNA parent strand.
Single stranded binding(SSB) proteins	Bind to ssDNA and prevent the DNA double helix from re-annealing after DNA helicase unwinds it, this maintaining the strand separation.
Topoisomerase	Relaxes the DNA from its super-coiled nature.
DNA Gyrase	Relieves strain of unwinding by DNA Helicases; this is a specific types of Topoisomerse.
DNA Ligase	RE-anneals the semi conservative strands and joins Okazaki fragments of the lagging strand.
Primase	Provides a starting point of RNA (or DNA) for DNA polymerase to begin synthesis of new strand.
Telomerase	Lengthens telomeric DNA by adding repetitive nucleotide sequences to the ends of eukaryotic chromosomes.

